Italian Named Entity Recognition with Transformer Models

UmBERTo vs XLM-RoBERTa on NERMuD 2023

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The question

Does language-specific pretraining beat multilingual pretraining for Italian NER?

PER
Person
Pope Leo

ORG
Organization
Agostino Gemelli Hospital

LOC
Location
Rome

The two models

UmBERTo
Italian-specific

32k	vocabulary
Italian	OSCAR corpus
WWM	whole word masking

XLM-RoBERTa
Multilingual

250k	vocabulary
100	languages, 2.5 TB
MLM	masked LM only

Dataset: NERMuD 2023

Wikinews

News articles

10,912

training examples

Fiction

Literary excerpts

11,423

training examples

ADG

De Gasperi speeches

5,147

training examples

27,482 train · 4,767 dev · 4,126 test Tags: B-/I- PER · ORG · LOC · O

Same training protocol

2e-5	8	3	256
learning rate	effective batch size	epochs	max sequence length

- Hugging Face Trainer, AdamW, seed 42
- Best checkpoint chosen on dev F1
- Evaluated with seqeval (entity-level P / R / F1)

Global result: UmBERTo wins

0.8945

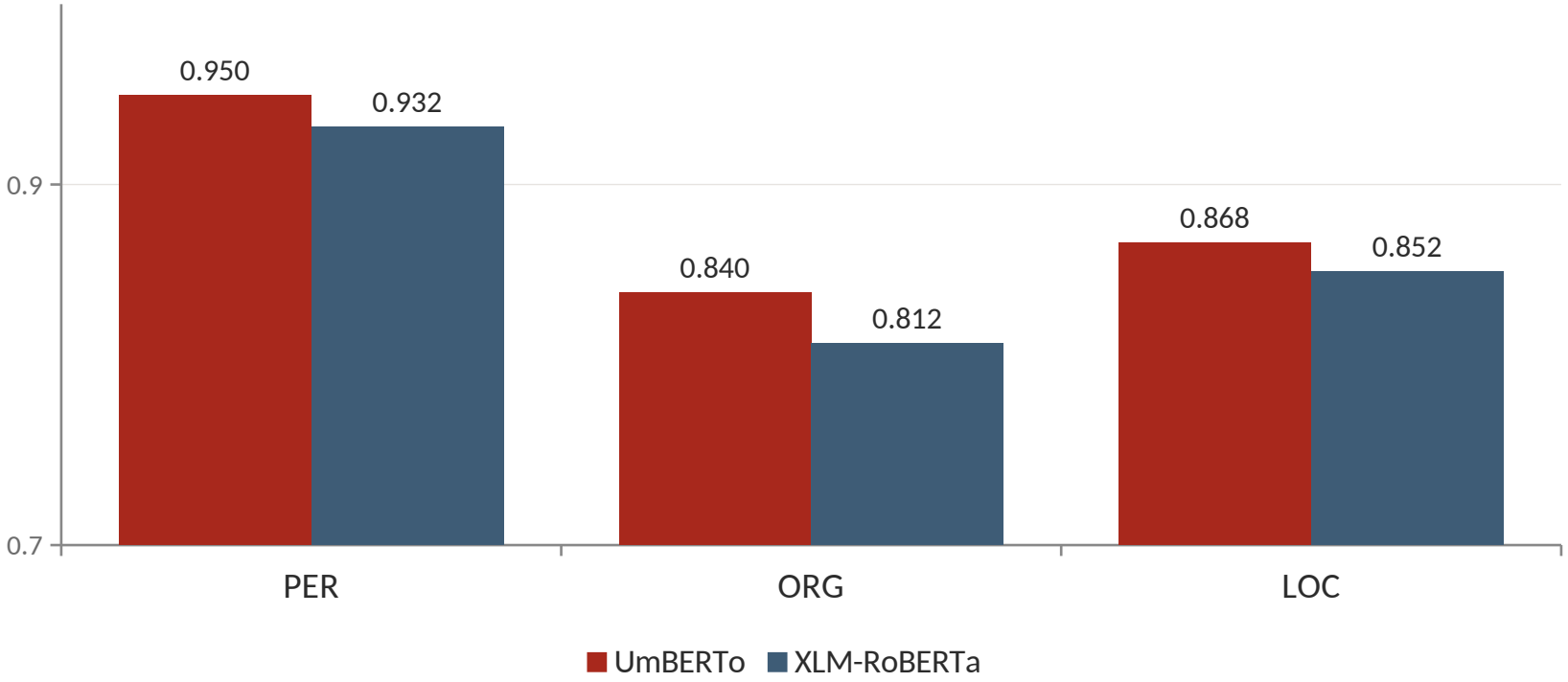
UmBERTo · test F1

0.8743

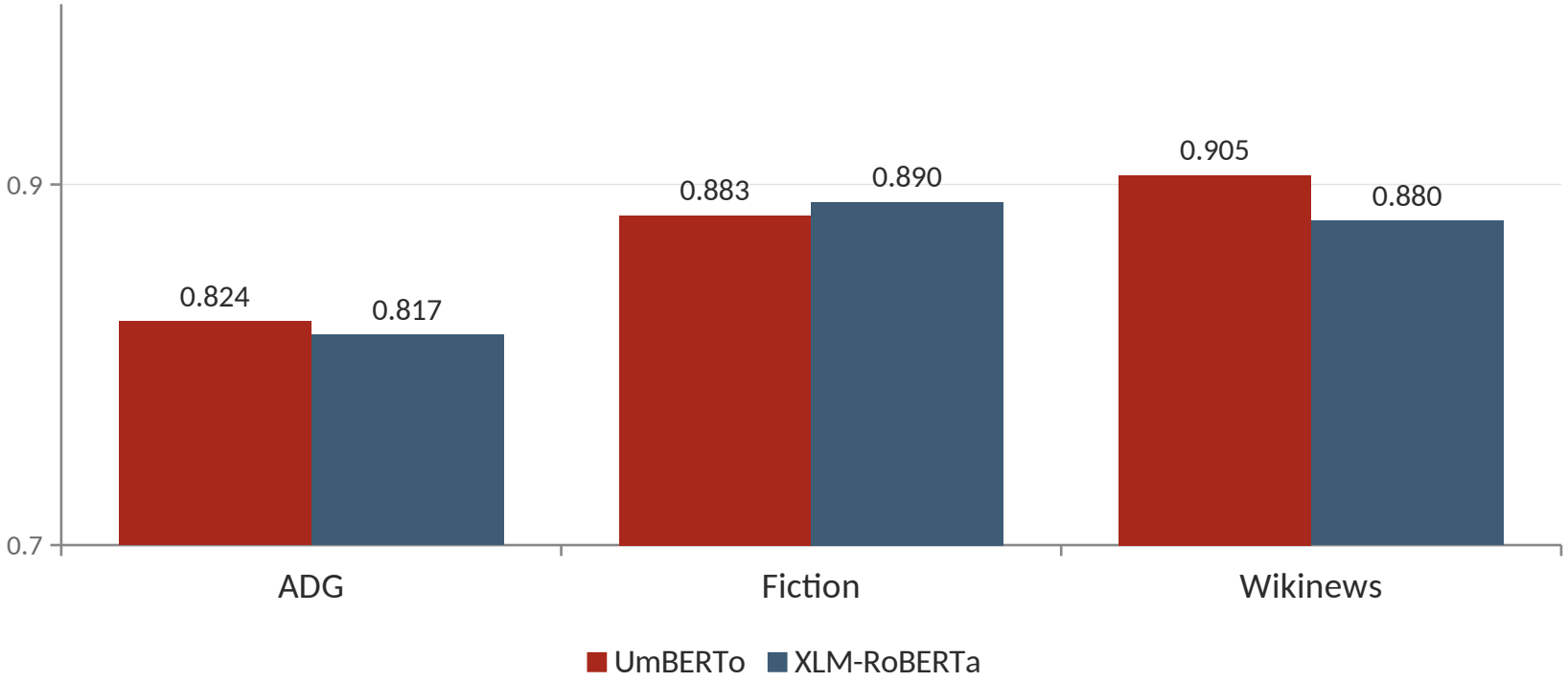
XLM-RoBERTa · test F1

Gap of ≈ 2 F1 points, consistent in precision and recall

By entity type



By domain


0.9
0.7
0.824
0.817
0.883
0.890
0.905
0.880
ADG
Fiction
Wikinews
UmBERTo
XLM-RoBERTa

Takeaways

What we found

- Italian-specific model wins overall
- Advantage holds for all entity types
- Fiction is the exception

Caveats & next steps

- Single run, no significance test
- Models differ beyond language
- Next: multiple seeds, larger models, other datasets